

Degree: B.Tech., Stream: ALL (Group A)

Term – I, First Semester Examination, August 2025

Subject Code: BSCM103A

Subject Name: Mathematics-Calculus

Full Marks: 30

Duration: 1 Hour

Part A (2 × 5)

Answer any 5 out of 6 questions

1. Verify Rolle's Theorem for the function $f(x) = x^2 - 5x + 6$ in $[1,4]$. 2
2. Using Lagrange's Mean Value Theorem, show that $0 < \frac{1}{x} \log \frac{e^x - 1}{x} < 1$, for $x > 0$. 2
3. Verify Cauchy's Mean Value Theorem for the functions e^x and e^{-x} in the interval $[a,b]$. 2
4. Find $\lim_{x \rightarrow 0} \frac{\tan x - x}{x - \sin x}$. 2
5. State Euler's theorem of 1st order for homogeneous functions. 2
6. Calculate the two repeated limits for the function $f(x, y) = \frac{2x+3y}{x-4y}$ as $x \rightarrow 0, y \rightarrow 0$. 2

Part B (5 × 2)

Answer any 2 out of 3 questions

7. Using Maclaurin's series, prove that 5

$$\log(1 + \sin x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{12} + \dots$$

8. Show that the function $f(x, y) = \begin{cases} \frac{(x^3+y^3)}{x-y}, & (x, y) \neq (0,0) \\ 0, & (x, y) = (0,0) \end{cases}$ is not continuous at $(0, 0)$. 5

9. If $u = \cos^{-1} \frac{x+y}{\sqrt{x}+\sqrt{y}}$ then prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0$. 5

Part C (10 × 1)
Answer any 1 out of 2 questions

10. Expand $\sin x$ in powers of $\left(x - \frac{\pi}{4}\right)$. Hence find the value of $\sin 46^\circ$ correct upto 4 places of decimal.

10

11. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, then prove that

(a) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{(x+y+z)}$

4

(b) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$

2

(c) $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = -\frac{3}{(x+y+z)^2}$

4

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